DADS 5250 · MODULE 7 HACKATHON · TOPIC BRIEF

Staffing Boston's 311 City Service Teams

Planning field crews for four City of Boston departments using real 311 service request data.

The problem	Decide how many crews four city departments need in each month of 2025.
Core difficulty	Seasonal demand, very different on-time performance across departments, and the cost and delay of hiring, releasing, and seasonal crews.
Data	Real City of Boston 311 requests for 2023 and 2024 (306,756 requests in 2024). Public, free to download. REAL
GenAI angle	A 311 intake assistant that turns a resident's plain-language message into the right request type and department (see Section 5).
Due	Sunday, October 25, 11:59 PM ET

1 Background

Boston 311 is the city's service request system. Residents report problems like potholes, missed trash pickups, broken streetlights, illegal parking, and rodents through the BOS:311 app, a phone call, or the website. City workers also log issues they spot in the field. In 2024, the city received 306,756 requests, about 840 a day.

Each request is routed to a city department and a team queue, and most come with a target date to close it. Demand is uneven. Public Works peaks in August and September, Parks more than triples between January and June, and requests vary by neighborhood and type.

Performance also varies a lot. In 2024, Public Works closed 90% of its requests on time, Parks 93%, and Inspectional Services 72%. Transportation, which handles parking enforcement, closed only 50% on time.

Your goal is to build a month-by-month staffing plan for 2025 that decides how many field crews four key departments need, using real 311 data from 2023 and 2024.

2 Key Rules & Workflow

The request data is REAL. Crew capacities, costs, and starting crew levels are planning assumptions ASSUMED given for this challenge, the same way a business case gives salaries. Use them as stated.

Departments in scope

Department	Code in data	2024 requests REAL	On time, 2024 REAL	Crew capacity ASSUMED	Annual cost per crew ASSUMED	Crews, Jan 2025 ASSUMED
Public Works	`PWDx`	157,610	90%	12 requests per day	$130,000	50
Transportation (parking & traffic)	`BTDT`	81,447	50%	15 requests per day	$110,000	18
Parks & Recreation	`PARK`	18,736	93%	6 requests per day	$120,000	10
Inspectional Services	`ISD`	18,624	72%	4 requests per day	$140,000	16

Request flow

- Each request is assigned to a department and a team queue, and most get a target close date (the `sla_target_dt` column).
- A request is on time if it closes by its target date, and overdue if it doesn't.
- You can assume no requests are waiting at the start of 2025.

Crew capacity & lifecycle

- A crew works 21 days a month and closes up to its daily capacity each day.
- New crews need 1 month of hiring and training before they start.
- Releasing a crew requires 1 month of notice and costs 25% of its annual cost.
- Parks may also hire seasonal crews from April to October at 1.3 times the monthly cost, with no training delay and no release cost.

Cost of an overdue request ASSUMED

- Each overdue request costs $20 in repeat calls, rework, and lost resident trust.
- A request still open 30 days after its target date costs an extra $100.

3 The Challenge: A Month-by-Month Staffing Plan for 2025

Decide how many crews each of the four departments needs for each month of 2025, balancing:

- **Cost efficiency:** avoid idle crews in slow months.
- **Timely service:** raise the share of requests closed on time, especially for Transportation.
- **Flexibility:** plan for seasonal swings given the hiring, release, and seasonal-crew rules.

Recommended deliverables

- **Demand forecast:** forecast monthly requests for each department in 2025, using 2023 and 2024 data.
- **311 intake assistant with GenAI:** the data has no free-text descriptions from residents. Build an assistant that reads a resident's plain-language message (for example, "there's a big hole on Centre St

near the T stop") and picks the right request type and department queue from the real list in the data. Create a small test set of messages and report how accurate it is. The `case_title` column has thousands of messy variants, so you can also use an LLM to clean and group them.

- **Dynamic staffing model:** a table showing crews by department for each month of 2025, including when you hire or release crews.
- **Analysis & justification:** a short report explaining your key decisions and assumptions, and their impact on cost and on-time performance.
- **Sensitivity analysis:** how your plan holds up if requests are 20% higher or lower than forecast, or if a snowstorm or heat wave causes a spike.
- **Other considerations:** any other factor you recommend considering, such as neighborhood equity in response times, or how requests from city employees differ from requests from residents.

4 Data Provided

All data is real and public, published by the City of Boston under a public domain license. Download it from the 311 Service Requests page on Analyze Boston (https://data.boston.gov/dataset/311-service-requests).

File	What it contains
311 Service Requests - 2024 (direct download, 174 MB CSV)	All 306,756 requests opened in 2024
311 Service Requests - 2023 (on the same page)	All requests opened in 2023

Each row is one request, with 30 columns. The most useful are:

Column(s)	Meaning
`case_enquiry_id`	The request ID
`open_dt`, `sla_target_dt`, `closed_dt`	When it was opened, when it was due, and when it was closed
`on_time`	ONTIME or OVERDUE
`case_status`, `closure_reason`	Open or Closed, and how it was closed
`case_title`, `type`, `reason`, `subject`	What the request is about, from most to least specific (170 types in 2024)
`department`, `queue`	Who handled it (for example `PWDx` and `PWDx_Code Enforcement`)
`source`	How it came in: Employee Generated, Citizens Connect App, Constituent Call, City Worker App, or Self Service
`neighborhood`, `location`, `latitude`, `longitude`	Where it is, plus ward and district columns

Things to watch for in the real data

- About 16% of 2024 requests have no target date.

- Some target dates are very long. For example, many Parks requests have a target of 365 days.
- 42% of requests were logged by city employees, not residents.
- `case_title` is messy, with about 3,700 different values, including internal codes. Some text contains HTML codes like `&`.
- Timestamps are in local Boston time.
- If you want to compare your plan with what actually happened in 2025, note that the city switched to a new 311 system in October 2025. The 2025 data is split across the "2025" file and the "NEW SYSTEM" file, which have different columns.

5 Making It a GenAI Application

The hackathon asks you to build a working AI application that combines **at least two course techniques** (API calls, prompt engineering, structured output, LangChain, RAG, or agents). The planning model above is the problem; your AI application is how you solve it and put it in people's hands. The directions below are examples, not requirements. Pick one, combine several, or propose your own.

AI ASSISTANT

311 Intake Assistant *(the core GenAI deliverable)*

A resident describes a problem in plain English. The assistant retrieves the closest of the 170 real request types, returns the type, department, and queue as structured output, and asks a follow-up question when the message is unclear.

Techniques: RAG, structured output, prompt engineering

CONTENT PIPELINE

Department Performance Reporter

Turns raw monthly 311 data into a written report per department or neighborhood: volumes, on-time rate, trends, and recommended staffing changes from your model.

Techniques: LangChain chains, prompt engineering, structured output

DATA ANALYZER

Ask Boston 311

Ask questions in plain English ("Which neighborhoods wait longest for pothole repairs?") and get charts and written insights from the 311 data.

Techniques: Agents (code or tool use), structured output

WORKFLOW AUTOMATOR

Intake-to-Plan Pipeline

Classify a stream of incoming messages, route them, estimate each department's load and on-time risk for the coming month, and flag where crews are short.

Techniques: Agents, structured output, LangChain

6 Scoping for One Week

- Limit your analysis to the four departments in scope and aggregate to monthly counts early. The full file is large.
- Build a short, de-duplicated list of request types for your intake assistant before you start prompting.
- Write a test set of around 30 to 50 realistic resident messages with the correct answers so you can measure accuracy.
- The best hackathon projects are not the most complex. They are the most focused. A small model that works end to end beats a big one that doesn't run.

7 What to Submit

Due **Sunday, October 25, 11:59 PM ET** on Canvas (7 — Hackathon Project). This is a group assignment: one team member submits, and all members contribute to every deliverable.

- Your completed Jupyter notebook (`.ipynb`) with every cell run and all outputs visible.
- Your GitHub repo link, containing the code and a README with setup instructions, an architecture diagram, and each team member's contributions.
- A link to your 3-minute demo video walking through your product, your architecture, and the part you're most proud of.
- Review the rubric attached to the assignment to make sure your submission meets every requirement.

AI use. You may use AI tools, but explain how you used them and clearly mark AI-generated sections. You are responsible for making sure everything you submit is accurate and follows Northeastern's Academic Integrity and Use of AI policy (linked in the syllabus).

8 Choosing This Topic

Each team works on one topic. Topics are assigned **first come, first served** through the topic sign-up sheet, and each topic has a limited number of team slots. Check that this topic still has open slots, then register your team. Don't start building until your topic is confirmed on the sheet.

Questions about the topic? Post them in the Discussions Q&A Forum on Canvas so other teams can benefit, or bring them to office hours.